On the Dual Origin of Ontological Deltas: Spectral Necessity vs. Axiomatic Minimality in the TNA–ISSP Framework

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Abstract

We clarify the distinct yet complementary origins of *ontological deltas*—the minimal structural supports that sustain coherence in quantum and classical systems—within two foundational principles: the **Theory of Axiomatic Necessity (TNA)** and the **Induced Spectral Structure Principle (ISSP)**. While both frameworks predict the existence of discrete, necessary anchors that stabilize physical states, they do so from fundamentally different premises: TNA from **logical minimality under reduction to incoherence**, ISSP from **differential geometry**.

Crucially, the TNA v1.2 Addendum establishes that these deltas are **not axioms in the conventional logical sense**, but **irreducible external constraints whose removal collapses the system into functional incoherence**. They are **detectable, physical, and non-derivable from within the operational domain (N0)**.

In contrast, ISSP shows that in any first-order differentiable system, such deltas **necessarily emerge as eigenvectors of an induced spectral metric**. Thus, while ISSP implies spectral deltas universally in continuum systems, TNA posits structural deltas even in non-differentiable domains. In physical systems, the two converge: TNA's irreducible supports are realized as ISSP's eigenmodes. This asymmetry establishes ISSP as the deeper physical principle for continuum systems, while TNA provides a more general ontological scaffold across all domains of coherent structure.

Keywords: ontological deltas, spectral theory, boundary-induced collapse, TNA, ISSP, structural necessity, quantum measurement, coherence, minimal support.

1. Introduction

The quest to understand why physical systems manifest definite outcomes—rather than indefinite superpositions—has led to numerous interpretations of quantum mechanics. Most remain instrumental: they predict statistics but offer no mechanism for *outcome uniqueness*. The **Boundary-Induced Collapse (BIC)** framework, grounded in the **Theory of Axiomatic Necessity (TNA)**, resolves this by asserting that collapse arises from *structural minimality*: only a finite set of external constraints (the "deltas") can sustain coherence. Separately, the **Induced Spectral Structure Principle (ISSP)** demonstrates that any first-order differentiable system necessarily generates a symmetric spectral metric whose eigenvectors define stable modes.

Both frameworks predict *discrete anchoring points*—deltas—but from radically different starting points. This paper clarifies their relationship, domain of validity, and ontological hierarchy.

2. Deltas in the Theory of Axiomatic Necessity (TNA)

2.1 Logical Minimality as Ontological Ground (post-Addendum)

The TNA v1.2 Addendum refines the original theorem by replacing the language of "axioms" with that of **structural necessity under reduction to incoherencia**. It proves that any coherent system SS operating in the observable domain **N0** requires a **finite, irreducible core of external constraints**—denoted F_{min} such that:

- $F_{\text{min}} \not\subseteq N_0$ (exteriority),
- $S \cup F_{\text{min}} \models E$ (sufficiency),
- $\forall f \in F_{\text{min}},\ (S \cup (F_{\text{min}} \setminus \{f\})) \not\models E$ (irreducibility).

Here, E represents the set of observable sentences (e.g., stable coherence). The removal of any element f triggers **reduction to incoherencia**—not logical contradiction, but **functional collapse**.

These elements of F_{min} are the **ontological deltas**. They are **not assumed**; they are **revealed by their absence**. In physical instantiations (e.g., quantum devices), they manifest as **discrete, experimentally locatable points** in boundary-parameter space—precisely the "bolts" of your engineering metaphor.

> ***Key refinement****: Deltas are* ***not axioms****; they are* ***necessary external supports whose omission is empirically falsifiable via loss of coherence****.*

2.2 No Requirement of Spectral Structure

TNA applies equally to discrete, symbolic, or non-differentiable systems (e.g., legal frameworks, cellular automata). Thus:

- **TNA ⇒ deltas**,
- **But TNA ⇏ spectral structure**.

This makes TNA a **universal theory of coherence**, while ISSP governs its **physical realization**.

3. Deltas in the Induced Spectral Structure Principle (ISSP)

3.1 Spectrality as a Consequence of Differentiability

ISSP begins with a differentiable mapping
$f : \mathbb{R}^n \to \mathbb{R}^m$
with Jacobian J.

It defines the induced metric

$$G = J^\top J,$$

which is symmetric and positive definite (if J has full rank).

By the spectral theorem, G admits an orthonormal eigenbasis
$\{v_i\}$ with eigenvalues $\{\lambda_i > 0\}$.

These eigenvectors are **structurally stable modes** — the only directions in which the system can respond coherently to perturbations.

In physical terms, they are the **spectral deltas**.

> ***Key property***: *ISSP deltas are **mathematically derived**, not postulated. They emerge inevitably from differentiability + inner product.*

3.2 Universality Within Differentiable Systems

ISSP applies to:

- Linear systems (normal equations),
- Quantum measurement (Hermitian observables),
- Control theory (Lyapunov matrices),
- Statistics (Fisher information),
- Mechanics (inertia tensors).

In all cases, the eigenvectors are the **only structurally admissible states**. Collapse, optimization, or resonance occurs along these axes.

> ***ISSP ⇒ spectral deltas,***
> ***And ISSP ⇏ TNA*** *(though it is compatible).*

4. The Asymmetry of Implication

We now state the central result:

> ***ISSP implies the existence of spectral deltas in all first-order differentiable systems.***
> ***TNA implies the existence of axiomatic deltas in all coherent systems, differentiable or not.***
> ***Therefore, spectral deltas are a *subset* of possible ontological deltas.***

In other words:

- **All spectral deltas are ontological deltas** (they are necessary for structural stability).
- **Not all ontological deltas are spectral** (e.g., in a discrete logical system with no derivative, deltas exist but have no spectrum).

This establishes a **hierarchical relationship**:

$$\text{TNA (general)} \supset \text{ISSP (physical / differentiable)}.$$

Yet, within the domain of physics — where systems are differentiable and embedded in geometric space — **ISSP provides the *mechanism* by which TNA's axioms are physically realized**.

5. Physical Unification: Boundary-Induced Collapse

In quantum devices, the **boundary operator** B encodes experimental and environmental constraints. Within BIC:

- **TNA view:** B defines the minimal set N_1; missing any element causes decoherence.
- **ISSP view:** B induces a metric $G = B^\dagger B$; its eigenvectors are the only admissible collapse states.

Thus, in physical systems:

$$\text{TNA delta} \equiv \text{ISSP eigenvector}$$

$$\text{TNA weight} \equiv \text{ISSP eigenvalue}$$

The **realizability functional** $F(\rho, B)$ is minimized precisely when ρ aligns with a dominant eigenvector of $B^\dagger B$. Collapse is not random — it is **spectral selection under axiomatic constraint**.

6. Domain-Specific Applicability

DOMAIN	TNA APPLICABLE?	ISSP APPLICABLE?	PRIMARY DELTA TYPE
Quantum devices	✅	✅	Spectral (ISSP) = Axiomatic (TNA)
Classical mechanics	✅	✅	Spectral (inertia, stiffness)
Statistical inference	✅	✅	Fisher information eigenvectors
Symbolic logic	✅	❌	Purely axiomatic
Cellular automata	✅	❌	Discrete necessity anchors
Economic equilibria	✅	⚠️ (if differentiable)	Context-dependent

This table shows that **TNA is the broader ontological framework**, while **ISSP is the physical instantiation principle** for continuum systems.

7. Conclusion

The ontological deltas predicted by TNA and ISSP are not competing concepts, but **complementary layers of necessity**:

- **TNA** answers *why* coherence requires discrete supports: because coherence is logically minimal.
- **ISSP** answers *how* those supports manifest in physical reality: as eigenvectors of an induced spectral metric.

In differentiable systems, the two converge seamlessly. But **only ISSP explains *why* the deltas have spectral form**; **only TNA explains *why* deltas must exist even when no spectrum is definable**.

Thus, **spectrality is a sufficient condition for delta realization in physics, but not a necessary condition for delta existence in ontology**.

This dual foundation restores **causal clarity** to quantum measurement: collapse is neither stochastic nor mental, but the **structural consequence of boundary-enforced spectral necessity**, grounded in a deeper logic of minimal coherence.

> ***Final statement***:
> *Reality does not choose outcomes. Boundaries forbid alternatives—and in differentiable worlds, they do so through the language of spectra.*

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